

PIA 409 Course Outline

PIA 409 – Climate Risk in Project Appraisal

Expected annual loss, adaptation benefit, and the gate that stays PENDING.

School	04 – Public Investment
Hours	30 (indicative band 27-33)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 404 for the ENPV that climate risk reduces, PIA 406 for the carbon price this course finds unused, and PIA 407 for the switching value the Resilience Rating System's top grade requires.
Audience	Appraisal economists who have to produce or review a climate assessment, and the officials who have to decide what a risk rating means
Pass mark	60 %

What this course is

Climate enters an appraisal in two quite different places, and they are routinely confused. Physical risk reduces the value of the asset being built: a road that floods delivers fewer benefits and costs more to maintain. Greenhouse gas emissions are an externality the project imposes on everybody else. The first changes the ENPV of this project; the second changes the world's.

The register holds 7 climate assessments across 11 projects, 11 hazard screenings, and 11 environmental impact records. Expected annual loss is computed as annual probability times exposed value times vulnerability, discounted over the operating years – arithmetic simple enough that every figure in this course can be checked by hand, and this course checks all of them.

Two findings drive the material. The hazard screenings are recorded under SSP2-4.5 and the loss calculations under historical, so the climate-adjusted ENPV is a historical-climate number wearing a forward-scenario label. And 94332 tonnes of carbon dioxide equivalent are recorded across the register with a governed price of UTD 0 bn available, and monetised on 0 of 11 records.

What a graduate can do

Distinguish physical climate risk from an emissions externality and say where each enters an appraisal; compute an expected annual loss from probability, exposure and vulnerability and check a stored one;

discount a loss stream and reconcile a present value of expected loss by hand; read a climate risk index and say what it does and does not measure; explain why a risk index and an economic loss can rank projects differently; name a climate scenario and state what an unlabelled hazard estimate is worth; appraise an adaptation measure as an investment in its own right; monetise an emissions quantity against a governed carbon price; and state what a Resilience Rating System grade would require.

What this course does not claim

This course does not produce climate projections. The hazard probabilities in the register record their source as “SAMPLE_DATA hazard screening” and the climate integration is configured to a local model with a mock fixture. Where a probability comes from is a question for the climate modelling school; what an appraisal does with it is the question here.

It does not resolve the scenario question. Choosing between SSP1-2.6 and SSP5-8.5 is a policy decision with a defence, and this course teaches how to state which was used and why – not which to pick.

It does not value biodiversity, air quality or water quality. The environmental records here carry greenhouse gases only, and the course says so rather than implying the environmental case is complete.

It does not assign a Resilience Rating. The rating is a published external methodology; the course teaches what its grades require and shows that PIAS already has the machinery for the hardest of them.

Learning outcomes

1. Distinguish physical climate risk from an emissions externality.
2. Compute an expected annual loss from probability, exposure and vulnerability.
3. Discount a loss stream and reconcile a stored present value by hand.
4. Compute a climate-adjusted ENPV and express the reduction.
5. Read a climate risk index and say what it measures.
6. Explain why a risk index and an economic loss rank projects differently.
7. Name a climate scenario and state what an unlabelled estimate is worth.
8. Identify a scenario inconsistency between two tables.
9. Appraise an adaptation measure as an investment in its own right.
10. Compute the avoided loss that an adaptation measure buys.
11. Monetise an emissions quantity against a governed carbon price.
12. State what each Resilience Rating System grade would require.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	Two different things called climate	1	1.25	–	1	Quiz M01
M02	Expected annual loss	1	3.25	Lab 01	2	Quiz M02

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M03	The loss stream and its present value	1	1.50	–	3	Quiz M03
M04	The climate-adjusted ENPV	1	3.50	Lab 02	4	Quiz M04
M05	The risk index, and what it measures	1	1.50	–	5	Quiz M05
M06	Two rankings that disagree	1	3.50	Lab 03	6	Quiz M06
M07	Naming the scenario	1	1.50	–	7	Quiz M07
M08	The mismatch in this register	1	1.50	–	8	Quiz M08
M09	Appraising an adaptation	1	3.50	Lab 04	9	Quiz M09
M10	The avoided loss	1	1.50	–	10	Quiz M10
M11	Emissions, and the price that exists	1	3.50	Lab 05	11	Quiz M11
M12	Writing the climate section	1	1.50	–	12	Quiz M12

Laboratories

Lab	Title	Hours	Deliverable
1	Recompute every expected annual loss, by hand	2.0	A verified hazard table
2	Reconcile the present value of loss, and adjust the ENPV	2.0	A reconciled climate adjustment
3	Rank the portfolio twice, and triage it	2.0	A two-way ranking and a triage recommendation

Lab	Title	Hours	Deliverable
4	Appraise an adaptation measure, and test it against a second scenario	2.0	An appraised adaptation measure
5	Monetise the emissions, and name what is missing	2.0	A monetised emissions account

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The hazard screening, checked	25	Marked on the recomputation and on identifying the intensity gap
A2	The adjustment, reconciled	25	Marked on the reconciliation and on what the subtraction misses
A3	The scenario, and the mismatch	25	Marked on demonstrating that the numbers are the same
A4	Adaptation and emissions	25	Marked on avoiding double counting and on naming what is missing

Final project

A complete climate section for one project: the hazard screening with probability, exposure and vulnerability for each hazard and the expected annual loss recomputed by hand; the loss stream discounted and reconciled against the stored present value; the climate-adjusted ENPV with the reduction expressed as a percentage; the risk index with a statement of what it measures and how its ranking differs from the ranking by economic loss; the scenario named on every figure, with the mismatch in the register identified; an adaptation measure appraised as an investment; the project's emissions monetised against the governed carbon price; and a closing paragraph stating which Resilience Rating grade the evidence would support.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.